

- 1 The Poiseuille equation relating the volume flow rate $\frac{V}{t}$ of a fluid under laminar conditions through a horizontal tube of length L and internal radius r is

$$\frac{V}{t} = \frac{\pi p r^4}{8\eta L}$$

where p is the pressure difference between the two ends of the tube and η is the viscosity of the fluid.

- (a) Show that the SI base units for η is $\text{kg m}^{-1} \text{s}^{-1}$.

$$\begin{aligned} [\eta] &= \frac{[p][r]^4}{[L]\left[\frac{V}{t}\right]} \\ &= \frac{\text{kg m}^{-1} \text{s}^{-2} \cdot \text{m}^4}{\text{m} \cdot \text{m}^3 \text{s}^{-1}} \\ &= \text{kg m}^{-1} \text{s}^{-1} \text{ (shown)} \end{aligned}$$

[2]

- (b) In an experiment to determine η for water, a student recorded the following measurements in SI units, as shown in Table 1.1.

Table 1.1

quantity	magnitude in SI units	percentage uncertainty / %
$\frac{V}{t}$	1.0×10^{-6}	3
p	500	2
L	0.20	0.5

The internal diameter of the tube was measured and recorded as (0.200 ± 0.002) cm.

- (i) Calculate the percentage uncertainty in the internal radius r of the tube.

$$\begin{aligned} d &= 2r \\ \frac{\Delta d}{d} &= \frac{\Delta r}{r} \\ \% \text{ uncertainty} &= \frac{0.002}{0.200} \times 100\% \\ &= 1\% \text{ or } 1.0\% \text{ (1 or 2 s.f.) [A1]} \end{aligned}$$

percentage uncertainty = % [1]

- (ii) Using the results in Table 1.1 and (b)(i), determine η with its associated uncertainty. Give your answer to an appropriate number of significant figures.

$$\eta = \frac{\pi p r^4}{8L \left(\frac{V}{t} \right)}$$

$$= \frac{\pi \times 500 \times (0.10 \times 10^{-2})^4}{8 \times 0.20 \times 1.0 \times 10^{-6}}$$

$$= 9.817 \times 10^{-4} \text{ kg m}^{-1} \text{ s}^{-1} \text{ [C1]}$$

$$\frac{\Delta \eta}{\eta} = \frac{\Delta p}{p} + 4 \frac{\Delta r}{r} + \frac{\Delta L}{L} + \frac{\Delta V/t}{V/t}$$

$$\frac{\Delta \eta}{\eta} = 0.02 + 4(0.01) + 0.005 + 0.03 \text{ [M1]}$$

$$= 0.095$$

$$\Delta \eta = 0.095 \times 9.817 \times 10^{-4}$$

$$= 9.326 \times 10^{-5}$$

$$= 9 \times 10^{-5} \text{ kg m}^{-1} \text{ s}^{-1} \text{ (1 s.f.) [A1] recognise } \Delta \eta \text{ to 1 s.f.}$$

$$\eta \pm \Delta \eta = (9.8 \pm 0.9) \times 10^{-4} \text{ kg m}^{-1} \text{ s}^{-1} \text{ [A1] recognise } \eta \text{ to same d.p. as } \Delta \eta$$

$$\eta = \dots \pm \dots \text{ kg m}^{-1} \text{ s}^{-1} \text{ [4]}$$

- (iii) State and explain which measured quantity has the greatest contribution to the uncertainty of η .

Internal diameter or radius, because of the power of 4, so greatest contribution to the uncertainty of viscosity [B1]

.....[1]

[Total: 8]

2 Two atoms X and Y have masses $2m$ and $3m$ respectively. The 2 atoms move head on towards